Driving in Sun Glare: Can SPADs help?

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How many of you have driven west during a sunset? The sky is pretty, but you almost go blind from the sun glare. AA reports that sun glare is part of nearly 3000 accidents annually that occur in the UK[7]. An analysis of crash-intersection data finds that the odds of glare related crashes are higher in east- and west bound traffic during sunrise and sunset compared to north- and south-bound traffic[2].

Turns out this is a problem for autonomous cars as well. In 2016, a Tesla model S collided with a white trailer, resulting in a fatality. The trailer made an unprotected left turn in front of the car against a brightly lit sky and "neither Autopilot nor the driver noticed the white side of the tractor trailer".[8]

Sun glare is usually direct, where it shines from the sun to the sensor, or indirect, where it is reflected off of surfaces like skyscrapers or wet roads. In either case, parts of the image are overexposed or corrupted by artifacts like halos due to lens flare[10].

Advancements in modern CMOS + HDR achieve a high dynamic range; but can passive SPAD imaging be an alternative sensing method to reduce saturation?

1 Defining Glare

Ideally in a digital camera, a point source in focus should contribute light to a single pixel, but in reality it contributes some intensity to the nearby pixels depending on the distance and lens architecture. This 2D function is known as the glare spread function of the lens (GSF). The center of the GSF is in the direct path of the light[6].

We have a point source that is transmitted by a lens to a sensor. Neither the optics nor the sensor is perfect. Aberrations in the lens cause veiling glare and lens flare. Veiling glare is due to scattering in the lens and reflections off the lens and the sensor. Sharp image artifacts are called lens flare - such as a ghost image of the lens aperture.

Consider a hypothetical system where the optics are perfect - there is no veiling glare or lens flare. The sensor is not ideal though. If the light source is too bright, the pixels saturate.

Here, I discuss ways to improve saturated images due to sun-glare for automotive tasks.

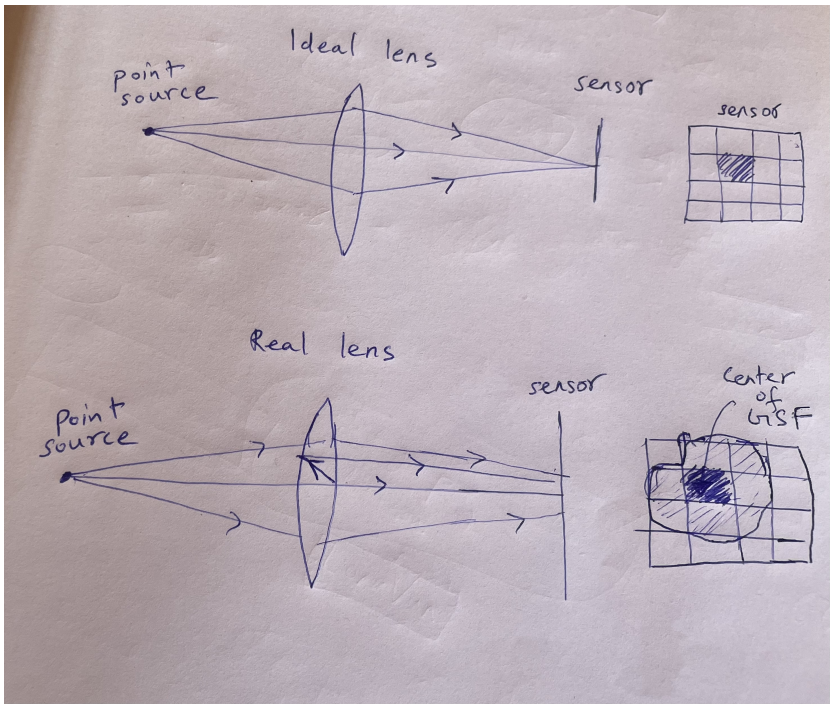


Figure 1: Glare spread function. Note how reflection off the lens surface causes glare.



Figure 2: A beach in sun glare. Notice how regions are overexposed. Shot on iPhone12

2 Saturation

Saturation is when an increase in incident light produces no response; the pixel has reached its maximum value. There is a loss of information in the scene since when the pixel is saturated, we only have a lower bound of the scene brightness[3].

2.1 CMOS

In a CMOS sensor, incident photons release electrons which are stored in a well. When the well capacity is reached, the sensor is saturated. The charge in the well is measured and then converted into 16 bit Analog to digital units (ADU). Errors in this conversion lead to read noise.

2.2 SPAD

A single photon avalanche diode is a type of photodetector that is able to detect individual photons and the time of photon arrivals to the order of picoseconds.

A passive free running SPAD (PF-SPAD) is used in applications where an active light source is not needed. It has a non linear response to intensity because of its dead time - where after detecting a photon, there is a time period during which the SPAD cannot detect anymore. A fraction of incident photons are missed (because of this dead time) and the response is non linear[11].

A PF-SPAD keeps track of the number of photon counts in a certain exposure time T.

Unlike a conventional CMOS sensor with a hard saturation limit, a PF-SPAD has soft saturation properties. Theoretically, it does not saturate even at high flux - the SPAD is still sensitive. At high flux, we are limited by the quantization noise and the dead time. In the limit, the PF-SPAD can detect a maximum of $\left\lceil \frac{T}{t_{dead}} \right\rceil$ photons. Photon counts are discrete and because of the

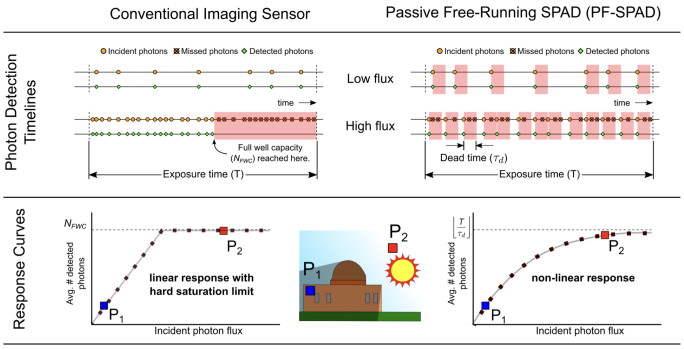


Figure 3: CMOS detects incident photons till it saturates and then it is not sensitive. With a SPAD, it is still sensitive at high flux, asymptotically reaching T/τ_{dead} at the limit. Image from Ingle et al. [11]

non linear saturation curve, even a ± 1 off from the true value leads to large change in the estimated flux value - we call this soft saturation but this occurs at a much higher flux value than conventional CMOS[11].

The dynamic range of a scene is defined as:

$$DR = 20 \log(E_{max}/E_{min}) \tag{1}$$

where E_{max} is the maximum measurable light intensity and E_{min} is the minimum [13].

3 Existing Methods

The presence of glare has tangible impacts on vision systems - increases false positives such as seeing traffic light or extra lane markings that are not there, and increases false negatives - not seeing a pedestrian, lane markings, etc. This leads to degrading downstream perception tasks such as detecting lanes, traffic signs and objects, degrading ADAS functions[5].

3.1 Saturation Aware Deconvolution

The observed glare ridden image is deconvolved into a glare free image using a pre-recorded glare spread function. But what happens to the GSF when the pixels are saturated? Alam et al account for saturation effects by estimating the true radiance in the saturated regions. This method was evaluated on downstream task performance like object detection and tracking, and performed better than methods that did not account for saturation [1].

The overexposed areas are reconstructed in post processing but what if the sensor captures the high photon flux?

3.2 HDR

Using HDR instead of auto exposure delivers improved performance for cars in direct sunlight, tested on lane detection algorithms [9] The most common HDR imaging technique is exposure bracketing, where multiple exposures are combined. But taking multiple exposures causes artifacts such as the LEDs in the traffic lights to flicker[4].

To address this, Yoo et al. propose a method to achieve a 120db dynamic range at 85 deg Celsius with a single exposure with a CMOS camera. Their method used two photodetectors, for low and bright light. Despite the high range, they report that when scaling, the small finite well capacity of the sensors leads to SNR degradation [14].

Advancements in CMOS + HDR clearly achieve a high dynamic range in a single exposure. But modern ADAS systems still struggle to respond in regions of high contrast. For instance, Tesla 2024+ model 3's driver assist manual mentions that the cameras are still prone to blindness in bright lights like direct sunlight and headlights[15]. If I were an automaker and I currently have CMOS cameras that get saturated in regions of high contrast, is it more viable for me to upgrade to an advanced CMOS or consider alternate sensing options like SPADs?

4 How could SPADs help?

Canon's 2025 SPAD camera uses "weighted photon counting" as opposed to counting every single photon, reducing power consumption by 75 percent whilst achieving a 156 Db dynamic range. Canon anticipates that this will be used in environments of contrast - such as a tunnel exit - or extremely low light [16].

How does a SPAD achieve such a high dynamic range?

We saw earlier that a PF-SPAD records the number of photons in a certain exposure time T.

Now consider an IP-SPAD, where the TDC (time to digital converter) keeps track of the counts and the picosecond resolution time stamps since the last photon arrival. The time between photon arrivals is the time of darkness.

The flux is calculated from this time of darkness. Notice how smaller periods of darkness is caused by high flux. This time measurement is inherently continuous, so its uncertainity comes from the floating point resolution and jitter in the electronics. This leads to an increased HDR when compared to a conventional CMOS and a PF-SPAD [12] This approach gives an extreme dynamic range in a single capture.

5 Direction(s) I'm interested in

Is this increase in dynamic range from SPADs needed to deal with sun glare saturation?

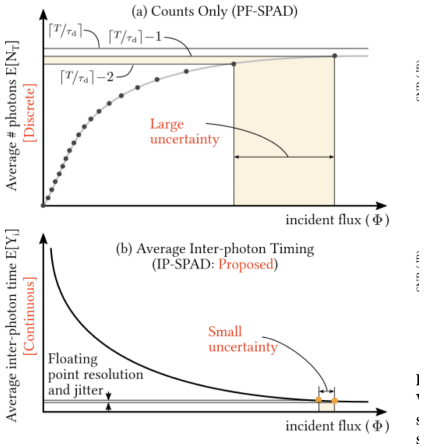


Figure 4: PF-SPAD vs IP-SPAD response. Image from Ingle et al. [12]

In what cases will using SPADs be preferred to advanced CMOS sensors?

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